Detecting departure and destination aerodromes from ADS-B

Version 4.5 — detection inside the pipeline

EUROCONTROL Performance Review Unit

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Summary

Versions 1–3 of this study ran outside the OPDI pipeline. They reimplemented aerodrome proximity with their own coordinate grid, which made every recommendation awkward to adopt: the findings were expressed in nautical miles while production ranks by H3 cell membership, so acting on them meant re-deriving rather than transcribing.

Version 4 moves detection **into the pipeline**. Three modes now share one candidate-detection stage — H3 as the index, exact great-circle distance as the comparison — and are selected by a `mode` parameter on the production function. The pipeline ran end to end on three days of 2025-06, producing a real flight list scored against Network Manager ground truth.

Three results:

Endpoint abstention buys +12.47 pp of departure coverage for -0.84 pp of accuracy. A real trade, not a free win — and a different conclusion from version 3, for a reason worth stating up front.

The detection radius barely matters; the height threshold is everything. Extending the radius from 40 NM to 110 NM changes departure coverage by +0.20 pp. Raising the height cut-off from on-ground-only to unrestricted changes it by more than fifty points. A distance cap costs almost nothing, which is worth knowing because it is the parameter most likely to be tuned.

Preferring aerodromes with scheduled service is close to free. Ranking candidates on a distance penalised for aerodromes without scheduled service gains **+0.67 pp of departure accuracy** and **+1.24 pp of arrival accuracy**, at no measurable cost in coverage.

! The production baseline improved too

Version 3 concluded that endpoint abstention beat production on *both* coverage and accuracy. That no longer holds, and the reason is not that abstention got worse — it is that **the baseline got better**. Running the altitude-trend algorithm on H3 candidates with exact distances lifts its departure accuracy from 97.45% in version 3 to 98.82% here.
The comparison in this version is therefore harder and fairer than the one before it. Version 3's numbers are not comparable to these and remain published unchanged.

i What changed since version 4

Every headline number is unchanged throughout. The point releases report more of what version 4 already computed, plus three new computations.
4.5

- A **result analysis** that picks an operating point and argues for it (Section 9), in flights rather than percentages, separately for departures and arrivals. It finds the currently published parameters are past break-even, and that no arrival setting beats production.
- **Why no single metric can choose** the parameters — each of coverage, accuracy and overall correctness is maximised at a degenerate corner (Section 9.3).
- **An approach cone tested against the rectangle** (Section 10.2), and what `endpoint` could borrow from `trend` for arrivals (Section 10).

4.3

- The height sweep runs to **40,000 ft** rather than stopping at 20,000. At 20,000 ft six points of departure coverage still separated the curve from the unrestricted case, so the old grid cut it off mid-rise and left the reader to guess where it flattened.

4.2

- A breakdown **by class of ground-truth aerodrome** (Section 8.1) — the one new computation. It surfaces the sharpest finding in the study: on small aerodromes and heliports the method is wrong **every single time it answers**, because the detection reference contains large and medium aerodromes only.
- Overall correctness is reported in the **radius and penalty sweeps**, not only in the results table.
- The per-aerodrome cross-check extends to the **100 busiest**, and now covers **arrivals as well as departures**.

4.1

- Results given for **both roles on all four measures**: arrival overall correctness and both in-area figures were computed but not printed.
- The sweep plotted **on accuracy as well as coverage**, for both roles. Accuracy peaks near the ground and falls as height is admitted, so the coverage view alone overstates the case for a high threshold (Section 5.1).
- The cross-check names the **mode and parameters** it was produced at, and separates the out-of-area bucket from the movement-count ratio it was distorting.
- Figure captions generated from the data rather than written by hand.

1 What the pipeline does before any method runs

The population a method is judged on is fixed long before the method executes. Each stage below discards something, and what survives is what detection sees.

1. Ingest (`ingestion/osn_statevectors.py`). Hourly partitions are read from the OpenSky archive and immediately filtered to the European bounding box `(-25.87, 26.75, 49.66, 70.26)` and decimated to one sample every 5 s. Both filters are applied *before* anything is written. For the three days here that turned roughly 60 GB of raw feed into **142,630,930 state vectors**.

What is discarded: everything outside Europe, and four samples in five. The 5 s decimation is the reason a track's "first sample" can be several seconds into the take-off roll rather than at the stand.

2. Track construction (`pipeline/tracks.py`). State vectors are grouped into tracks by `SHA2(icao24 || callsign)` and split on a gap of over 30 minutes, or over 15 minutes below 5,000 ft. Each sample is also indexed into H3 cells at resolutions 7 and 12.

What is discarded: nothing, but the grouping decides what counts as one flight. Version 2 measured the consequences — roughly one matched flight in ten is split across more than one track, and a few percent of tracks cover more than one flight.

3. Aerodrome zones (`reference/h3_airport_zones.py`). Each aerodrome is surrounded by concentric rings of H3 res-7 cells: 5 NM steps out to 40 NM, then 10 NM steps to 110 NM. Every cell carries the ring band it belongs to, so a consumer chooses its own radius at read time. **33,751,619 (aerodrome, cell) rows** for the 1,353 large and medium aerodromes in the area.

Why it reaches 110 NM: the same table serves ASMA ring crossings at 40 and 100 NM. Detection uses a fraction of it.

4. Endpoint candidates (`pipeline/flights.py`). For each track, the first and last sample are matched against the zone table on `h3_res_7`, and the exact great-circle distance to each candidate aerodrome is computed. The result is cached as `opdi_endpoint_candidates` — **7,491,655 rows**.

This is the step that makes the sweeps in this paper affordable. Deriving candidates from state vectors is expensive; once cached, changing a threshold is a filter and a re-rank over a few million rows rather than a pass over 142 million. Reducing to the two endpoint samples *before* the join is also what makes the full 110 NM reach cheap — about a million rows to match, not the whole month.

> **i** H3 as an index, distance as the comparison
>
> The H3 join answers "which aerodromes are plausibly near this sample" in one equi-join, without a spatial cross product. It does not answer "how near", because a cell inside a 40 NM ring can still be more than 40 NM from the aerodrome. So every candidate is then measured exactly, and the distance — not the cell — decides.

2 How the metrics work

This study turns on the difference between measures that are easy to conflate, so each is defined before any number appears.

Let G be the ground-truth flights and $A \subseteq G$ those the method answered.

Measure	Definition	The question it answers
Coverage	$\lvert A\rvert/\lvert G\rvert$	How often does it say *something*?
Accuracy	$\lvert \text{correct}\rvert/\lvert A\rvert$	When it speaks, how often is it right?
Overall correctness	$\lvert \text{correct}\rvert/\lvert G\rvert$	Of everything, how much did it get right?

A flight whose trajectory was never seen counts against coverage. That is deliberate: ground truth is the denominator throughout, so “we had no data” is a miss rather than an excluded row.

Accuracy alone is close to meaningless here. It says nothing about how often a method speaks, so a method that answers rarely and cautiously can post a better accuracy than one doing far more useful work. Consider two methods on the same flights:

Table 1: Accuracy favours the cautious method; overall correctness shows the eager one labels more than twice as many flights correctly.

Method	Coverage	Accuracy	Overall correctness
Cautious	40.00%	99.00%	39.60%
Eager	95.00%	88.00%	83.60%

Neither is simply better — they suit different uses. A flight list that downstream users treat as authoritative may well prefer the cautious one. The point is that **no single number decides it**, which is why all three are reported throughout.

2.1 Precision and recall, for out-of-area

Some flights have an origin or destination outside the observed area entirely. “Out of area” is therefore a legitimate *answer*, not a failure — and how well a method identifies those cases is measured with the standard pair:

Measure	Definition	The question it answers
Precision	of flights *labelled* out-of-area, the fraction that were	When it says out-of-area, is it right?
Recall	of flights that *were* out-of-area, the fraction it labelled	Of the cases out there, how many did it catch?

Precision is accuracy computed on one label instead of all of them. Recall has no counterpart in the table above, and it is the one that matters here: **high precision with low recall means a method is right when it fires but rarely fires**.

In the standard confusion-matrix vocabulary, with “out-of-area” as the positive class: precision is $TP/(TP + FP)$ and recall is $TP/(TP + FN)$. Coverage has no confusion-matrix equivalent, because it measures whether the method produced *any* label rather than which one.

2.2 In-area coverage and accuracy

Because roughly one flight in twelve is genuinely out of area, a method could raise its apparent accuracy simply by labelling out-of-area liberally. So the same two measures are also reported **restricted to flights whose truth is a real aerodrome** — detection skill with the free wins removed. When headline and in-area figures diverge, the gap is exactly the contribution of out-of-area labelling.

3 The three modes

All three read the same cached candidates. They differ in what evidence they consult and, more importantly, in when they are willing to stay silent.

3.1 `trend` — the current production algorithm

Evidence: every state vector below **FL40** that falls inside an aerodrome's detection zone, currently 30 NM.

Rule: for each (track, aerodrome) pair, smooth altitude over a 5-sample window, count how often the smoothed value rises versus falls, and require a margin of 4 either way. A positive margin is a take-off, negative a landing.

Abstains when: the margin is within ±4 — classified `ambiguous` and **the flight is dropped entirely**.

Failure mode: a flight seen only above FL40 near its origin, or with a noisy altitude trace, silently loses its aerodrome. The FL40 gate is the binding constraint, not the 30 NM radius.

3.2 `nearest` — the *traffic* library's rule

Evidence: the track's first and last sample. Nothing else — no altitude, no ground state, no trend.

Rule: name the nearest aerodrome by effective distance.

Abstains when: never, except when no candidate exists at all.

Failure mode: it answers just as confidently for a track that begins in the cruise as for one that begins at a stand. Its coverage is the highest of any method and its accuracy the lowest, and both facts have the same cause.

3.3 `endpoint` — nearest, with abstention

Evidence: the same two samples, plus their height above **field elevation** and their `on_ground` flag.

Rule: name the nearest aerodrome, but only if the endpoint is within `radius_nm` **and** either below `height_ft` above that aerodrome's elevation or flagged on the ground. Otherwise, if the endpoint lies near the boundary of the observed area, answer out-of-area. Otherwise answer nothing.

Why height above field elevation: a fixed altitude cut-off means nothing at an aerodrome sitting at 5,000 ft. Elevation comes from OurAirports and is missing for about 3% of large and medium aerodromes; where it is missing the height test is skipped and only the ground flag can satisfy the rule, rather than silently comparing against sea level.

Failure mode: it inherits `nearest`'s naming and therefore its confusions between neighbouring aerodromes; the abstention only decides *whether* to answer, never *what* to answer.

4 Results

Table 2: The three modes on 95,116 ground-truth flights, 2025-06-05 to 07. Overall correctness is coverage x accuracy. Produced at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty.

Movements	Mode	Coverage	Accuracy	Overall	In-area cov	In-area acc
departures	trend	68.72%	98.82%	67.91%	74.84%	98.95%
departures	nearest	91.68%	87.78%	80.48%	96.26%	91.17%
departures	endpoint	81.20%	97.98%	79.55%	87.43%	98.41%
arrivals	trend	68.06%	97.87%	66.61%	74.23%	97.94%
arrivals	nearest	88.56%	79.41%	70.33%	91.15%	84.20%
arrivals	endpoint	69.97%	95.10%	66.54%	75.89%	95.45%

`endpoint` sits between the two by construction, and that is the whole design: it keeps `nearest`'s naming rule and adds `trend`'s willingness to stay silent. It answers +12.47 pp more departures than production for -0.84 pp of accuracy.

Whether that trade is worth taking is a policy question rather than a technical one. A flight list treated as authoritative may prefer production's caution; an analysis needing volume may not.

Arrivals remain weaker than departures on every measure, as in all previous versions. A departure's first sample is at or near a stand, with the aircraft stationary shortly before. An arrival's last sample is more often a loss of reception during descent than a landing, and can be anywhere on the approach.

5 What a distance cap costs

The sweep varies the detection radius and the abstention height independently, 126 combinations in all, over the cached candidates. Height runs from on-ground-only to 40,000 ft above field elevation — past typical cruise, so the curve is followed to where it flattens — plus an unrestricted case as the limit.

Both measures are plotted for both roles, because the two axes do not merely differ in strength — they pull in opposite directions, and a coverage-only view hides that entirely.

Table 3: Detection radius at a fixed 15,000 ft abstention height. *ovr* is overall correctness, coverage x accuracy. Coverage saturates by about 30 NM for departures and 40 NM for arrivals; beyond that a wider radius trades accuracy for almost no coverage, and overall correctness flattens rather than turning – the radius has no interior optimum on this axis, unlike the height (Section 5.1).

Max radius	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
5 NM	71.62%	99.07%	70.95%	58.79%	98.64%	57.99%
10 NM	76.74%	98.77%	75.79%	62.36%	97.87%	61.03%
20 NM	80.07%	98.44%	78.82%	66.78%	96.65%	64.54%
30 NM	80.96%	98.16%	79.47%	69.10%	95.63%	66.08%
40 NM	81.20%	97.98%	79.55%	69.97%	95.10%	66.54%
60 NM	81.36%	97.79%	79.56%	70.29%	94.80%	66.63%

Max radius	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
80 NM	81.39%	97.75%	79.57%	70.38%	94.69%	66.64%
100 NM	81.40%	97.75%	79.57%	70.38%	94.67%	66.64%
110 NM	81.40%	97.75%	79.57%	70.38%	94.67%	66.64%

A cap costs almost nothing. Going from 40 NM to the full 110 NM reach changes departure coverage by +0.20 pp and accuracy by -0.23 pp. The useful range is 20–40 NM; below that coverage falls away quickly, above it nothing happens.

Every radius at or above 20 NM traces effectively the same coverage curve: at 5,000 ft they span 74.39% to 74.58%, about a point across a five-fold change. Only the 5 NM and 10 NM lines fall away, and they fall further as the height threshold rises, because a tight radius costs more once more of the sky is admitted.

Height is the parameter that matters. At 40 NM, departure coverage runs from 31.21% with the on-ground-only rule, through 67.05% at 2,000 ft, to 81.20% at 15,000 ft. That is over fifty points from one axis against a fraction of a point from the other.

Arrivals behave the same way with less room: from 36.01% to 69.97% over the same range. They also saturate later in radius — at 15,000 ft a 20 NM cap costs departures +1.32 pp of coverage but arrivals +3.61 pp, because an arrival's last recorded sample is further from the aerodrome than a departure's first.

Two endpoints of the sweep serve as sanity checks, and both behave as they should: at height 0 the rule reduces to "endpoint on the ground" and gives 31.21%, close to what a surface-samples-only method achieved in earlier versions; with height unrestricted and a wide radius it reaches 91.41%, close to `nearest`'s 91.68%, which is what it should converge to.

The curve does flatten, and the grid now reaches the flat part. At 40 NM the last measured height is within +0.06 pp of the unrestricted case for departures and +0.36 pp for arrivals, so a threshold at cruise level is indistinguishable from no threshold at all. The steepest remaining step is between 30,000 and 35,000 ft — +4.08 pp of arrival coverage in one 5,000 ft band, which is where first and last contact at cruise gets admitted.

5.1 What the coverage view hides

Read on accuracy, the same grid tells a different story, and the two together are what actually determines an operating point.

Accuracy peaks near the ground, then declines. At 40 NM, departure accuracy is highest at 2,000 ft (98.98%) and arrival accuracy at 500 ft (98.57%). By 15,000 ft they have given up -1.00 pp and -3.47 pp respectively. Every foot of extra height admits endpoints that are further into the climb or descent, and those are exactly the endpoints most likely to sit nearer a neighbouring aerodrome than the real one.

Arrivals pay roughly three times as much for the same height. That ratio is the single clearest statement of why this study has never recommended changing the arrivals algorithm: the same parameter that is nearly free for departures is expensive for arrivals, so one operating point cannot serve both well.

The radius matters for accuracy even where it does not for coverage. Above 40 NM a wider radius buys essentially no coverage, but it keeps costing accuracy: at 15,000 ft, going from 5 NM to 110 NM costs -1.32 pp of departure accuracy and -3.97 pp of arrival accuracy.

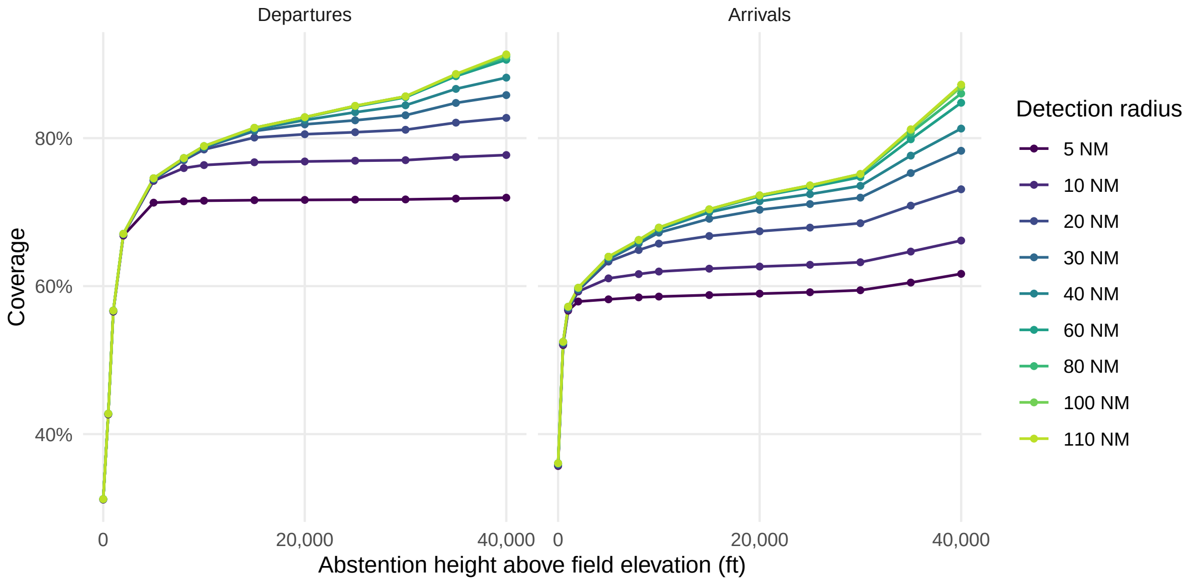

Figure 1: Coverage against abstention height, one line per detection radius. The shape of the curve belongs to the height axis in both panels; the radius only sets how much of it is reachable, and arrivals need more of it than departures. At 15,000 ft a 20 NM radius still leaves +1.32 pp of departure coverage and +3.61 pp of arrival coverage on the table against the full 110 NM reach, while at 40 NM both are within half a point of it.

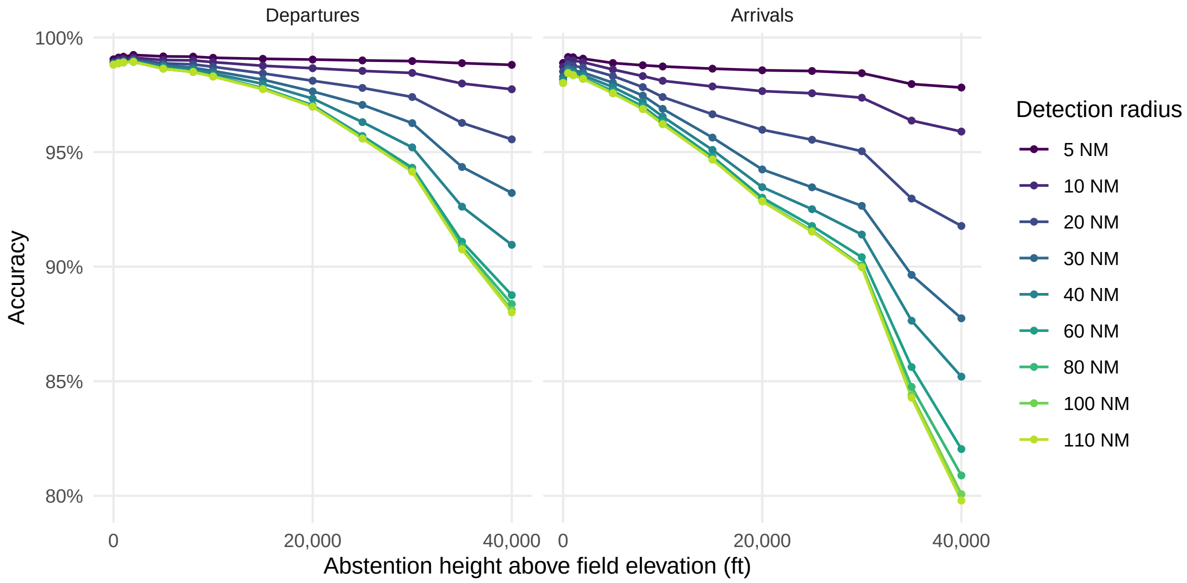

Figure 2: Accuracy against abstention height, same axes. This is the figure the coverage plot hides. Accuracy peaks close to the ground – at 40 NM, departures at 2,000 ft and arrivals at 500 ft – and every foot admitted above that is bought with correctness: -1.00 pp for departures by 15,000 ft, -3.47 pp for arrivals. The radius, which barely touches coverage above 40 NM, spreads accuracy at 15,000 ft by +1.32 pp for departures and +3.97 pp for arrivals.

This is the real argument for a cap. Version 4 justified one on the grounds that it costs almost no coverage; the accuracy view shows it is mildly *positive* rather than merely harmless.

Overall correctness is what reconciles them, and it has a maximum. Coverage rises steeply with height while accuracy falls gently, so their product rises for a long way: at 40 NM departure overall correctness runs from 66.36% at 2,000 ft to 79.55% at 15,000 ft. But it does not rise forever. It peaks at **25,000 ft** (80.41%) and declines above that, as the accuracy loss finally outruns the coverage gain.

> ⚠ Version 4.2 got this wrong, and the extended grid is why
>
> Version 4.2 stated that the product "keeps rising for departures across the whole range". That was true of the range it measured — the sweep stopped at 20,000 ft, below the turning point — and false of the parameter. Extending the grid to 40,000 ft was what exposed it. The claim is withdrawn.

The maximum is shallow, which is the practical point: 80.41% at its peak against 79.55% at the recommended 15,000 ft, a difference of +0.86 pp. Buying it costs -1.67 pp of accuracy, so the recommendation does not move.

Arrivals never turn over inside the grid: overall correctness is still climbing at 40,000 ft (69.26% against 66.54% at 15,000 ft). Arrival coverage is so far from its ceiling that admitting more of it still pays, which is the same asymmetry once more.

A user who cares about a trustworthy label rather than a complete one should ignore all of this and read the accuracy panel, where the optimum sits near the ground.

6 Preferring aerodromes with scheduled service

Version 3 found that the most frequent departure errors formed one pattern: a scheduled civil airport losing to an adjacent military airfield — Izmir to Çiğli Airbase, Naples to Grazzanise, Cagliari to Decimomannu, Catania to Sigonella, Gdańsk to Cewice, all between 7 and 25 NM apart.

The mechanism is straightforward. With 5 s sampling a departing aircraft's first recorded fix is already into the climb and displaced along the departure track. Where a military field lies between the airport and that track, it is genuinely the nearest aerodrome — the rule is not malfunctioning, it is answering the question it was asked.

Aviation knowledge resolves it without new data: a commercial flight almost certainly used the aerodrome with scheduled service. Candidates are therefore ranked on **effective distance** — true distance plus a fixed penalty for aerodromes without scheduled service — so a military field must be *clearly* nearest rather than merely nearest.

Table 4: Scheduled-service penalty at 40 NM and 15,000 ft. *ovr* is overall correctness. A penalty of 0 reproduces the unbiased ranking exactly and is the control.

Penalty	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
0 NM	81.20%	97.31%	79.02%	70.00%	93.85%	65.70%
5 NM	81.20%	97.72%	79.35%	69.99%	94.31%	66.01%
10 NM	81.20%	97.98%	79.55%	69.97%	95.10%	66.54%
15 NM	81.19%	97.90%	79.49%	69.92%	94.88%	66.35%
20 NM	81.20%	97.68%	79.32%	69.91%	94.80%	66.27%
30 NM	81.19%	97.59%	79.23%	69.82%	94.84%	66.22%

The gain is **+0.67 pp of departure accuracy** and **+1.24 pp of arrival accuracy**, for a coverage change of -0.01 pp. Overall correctness moves with it — +0.54 pp and +0.84 pp — which is what makes this the one parameter in the study with no trade attached.

Two features make this trustworthy rather than a lucky constant. The control at penalty 0 reproduces the unbiased ranking, so the effect is attributable. And the curve **peaks and turns** — accuracy is best around 10 NM and declines above it. A penalty that merely favoured larger aerodromes would improve monotonically; one that corrects a specific bias should overshoot, and this does.

> ⚠ It is not free — the cost is a mirror error
>
> Version 3 measured what this preference breaks: genuine movements at non-scheduled aerodromes get pulled to the nearby hub — Biggin Hill to London City, Northolt to Heathrow. In that analysis the mirror error became the **largest single class of remaining departure error**, at roughly 59% of them.
> The net is favourable, and the accuracy figures above include the cost. But a flight list built this way will systematically under-report movements at non-scheduled aerodromes, and anyone analysing general aviation or military traffic should know that before using it.

7 Out-of-area

A flight from Dubai to Amsterdam has a departure aerodrome that no European ADS-B feed can name: the aircraft entered the observed area already airborne. Scoring that as a detection failure understates every method. Treating "out of area" as an answer is an accounting correction, not a detection improvement.

7.1 How it is derived

On the ground-truth side, an aerodrome is relabelled `OOA` when its coordinates fall outside the ingestion bounding box, **or** when its ICAO code does not resolve in OurAirports. The unresolvable codes are overwhelmingly non-European, and a code that cannot be located is one no trajectory endpoint could have been matched to. This is what puts prediction and truth in the same alphabet.

On the prediction side, an endpoint within 30 NM of the bounding box edge is labelled out-of-area. The longitude margin is scaled by $1/\cos(\text{lat})$: a degree of longitude shrinks toward the pole, so an unscaled margin would be roughly twice as strict in northern Norway as in the Canaries.

Precedence is aerodrome first, border second. The opposite order looks equivalent and is not — version 3 found Ponta Delgada, which sits about 8 NM inside the western edge, having its departures labelled out-of-area with the aircraft still on the runway. Being near the boundary means "out of area" only when there is nothing better to say.

The border test carries no height condition. A flight leaving the observed area is at cruise level by definition, so any height cut-off would reject exactly the endpoints the branch exists to catch.

7.2 What it does to the numbers

Table 5: Out-of-area for departures. Only `endpoint` emits the label; the others never do, so their recall is zero by construction.

Mode	truly OOA	labelled OOA	OOA recall	OOA precision	headline acc	in-area acc
trend	8.30%	0.00%	0.00%	0.00%	98.82%	98.95%
nearest	8.30%	0.00%	0.00%	0.00%	87.78%	91.17%
endpoint	8.30%	0.74%	7.94%	89.44%	97.98%	98.41%

`truly OOA` is a property of the sample, not of the method, so it must be identical across all three rows — that it is confirms the decomposition is being computed correctly rather than being a per-method artefact.

The gap between headline and in-area accuracy shows how much out-of-area labelling contributes. It is small here, which is the honest reading: the mechanism is right but the geometric test catches only part of what is there. Version 3 measured this directly at **92–94% precision against 10–15% recall** — when the rule fires it is almost always correct, but it rarely fires. An inbound flight's first recorded sample appears wherever a receiver happens to catch it, which is usually well inside the boundary rather than at it.

The concept is right and the instrument is wrong. A phase-based test would fit the physics better: a track whose first sample is already in the cruise, neither near an aerodrome nor descending toward one, has an origin outside the observed area regardless of where it sits. That is the clearest single improvement available and is not implemented here.

8 Cross-check against the reference

Flight-level accuracy can hide errors that cancel: a departure misattributed from A to B leaves both counts wrong and the total right. Movement counts per aerodrome are what an operational user reads, so they are checked separately.

Table 6: Movement counts against Network Manager, from the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. A count ratio below 1 is an undercount. The second ratio drops the out-of-area bucket, whose own recall is poor (Section 7) and which otherwise carries the shortfall of every unlabelled flight.

Movements	Aerodromes	Count ratio	Excl. OOA	Median recall	Median recall, 50+ movements
departures	824	0.810	0.875	80.00%	97.35%
arrivals	805	0.695	0.756	60.00%	84.08%

8.1 By class of aerodrome

Every headline figure averages over populations that behave nothing like each other. The detection reference holds **large and medium aerodromes only**, so flights whose true aerodrome is a small field or a heliport cannot be named correctly at any radius or height. Grouping ground truth by its OurAirports class separates that from detection skill.

Table 7: Coverage and accuracy by the class of the *ground-truth* aerodrome, from the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Classes are OurAirports types. The detection reference contains large and medium aerodromes only.

Movements	Class	Aerodromes	Flights	Coverage	Accuracy	Overall
arrivals	large airport	331	80,683	76.11%	96.27%	73.27%
arrivals	out of area	1	7,964	5.12%	37.99%	1.95%
arrivals	medium airport	346	5,983	74.88%	89.55%	67.06%
arrivals	small airport	70	352	57.67%	0.00%	0.00%
arrivals	heliport	58	134	36.57%	0.00%	0.00%
departures	large airport	331	80,661	88.37%	99.14%	87.60%
departures	out of area	1	7,892	12.34%	64.37%	7.94%
departures	medium airport	352	6,069	78.22%	92.21%	72.12%
departures	small airport	76	361	49.58%	0.00%	0.00%
departures	heliport	65	133	39.10%	0.00%	0.00%

Small aerodromes and heliports are answered confidently and wrongly, every time. Accuracy on both classes is exactly zero, on 49.58% coverage for small airports and 39.10% for heliports. That is not a near-miss rate: the reference cannot name these aerodromes, so when the endpoint happens to fall within 40 NM of some large or medium field below 15,000 ft, the rule names that one instead. Across both roles, 483 of 980 such flights get an answer and none of them are right.

The absolute cost is small — under half a point of the sample — but the shape of the error matters more than its size. **These are not abstentions; they are false statements**, and nothing in the output distinguishes them from the 99.14% of correct large-airport departures. Two fixes are available and neither is implemented here: extend the reference to small aerodromes so they can be named, or make abstention aware of what the reference contains, so an endpoint nearest to an aerodrome the reference omits produces silence rather than its neighbour.

Between the classes the reference does hold, the difference is real but ordinary. Departures at large airports run 88.37% coverage at 99.14% accuracy, against 78.22% and 92.21% at medium ones. Since large airports are 84.80% of the sample, the headline figures are very close to the large-airport figures and should not be read as applying uniformly.

Table 8: The 100 busiest aerodromes by Network Manager departures, against the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Out-of-area is excluded – it is not an aerodrome and is reported separately in Section 7. *Count ratio* is OPDI movements divided by Network Manager movements, so it can sit near 1 even when the individual flights are wrong; *recall* is the share of Network Manager departures OPDI put at this aerodrome, and is the stricter measure.

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LTFM	2,282	2,211	0.97	96.49%
EHAM	2,148	2,165	1.01	99.77%
EDDF	2,094	2,087	1.00	99.43%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LFPG	2,084	2,071	0.99	98.85%
EGLL	2,001	2,024	1.01	99.35%
LEMD	1,773	1,761	0.99	99.21%
LEBL	1,591	1,582	0.99	99.25%
EDDM	1,573	1,567	1.00	99.43%
LIRF	1,449	1,445	1.00	99.65%
LTAI	1,435	1,387	0.97	96.17%
LEPA	1,417	1,401	0.99	98.80%
LGAV	1,306	1,280	0.98	95.94%
EGKK	1,223	1,218	1.00	99.43%
EIDW	1,185	1,166	0.98	98.14%
LSZH	1,178	1,168	0.99	98.81%
LOWW	1,156	1,142	0.99	98.10%
EKCH	1,140	1,133	0.99	99.30%
LTFJ	1,137	1,115	0.98	96.83%
LPPT	1,010	1,001	0.99	99.11%
LFPO	992	990	1.00	98.99%
LIMC	973	964	0.99	98.56%
ENGM	932	927	0.99	98.93%
EGCC	926	902	0.97	97.41%
EGSS	885	883	1.00	99.66%
EPWA	875	874	1.00	99.43%
EBBR	872	886	1.02	99.66%
EDDB	870	864	0.99	97.70%
LEMG	850	849	1.00	99.65%
ESSA	820	791	0.96	95.98%
EDDL	758	750	0.99	98.15%
LFMN	707	695	0.98	98.02%
LSGG	680	688	1.01	99.41%
LKPR	667	659	0.99	98.20%
EFHK	659	660	1.00	99.09%
LEAL	599	598	1.00	99.83%
LHBP	578	554	0.96	95.67%
EGPH	577	569	0.99	98.61%
EGGW	550	551	1.00	99.82%
EDDH	538	537	1.00	98.14%
GCLP	520	0	0.00	0.00%
EDDK	515	502	0.97	97.28%
LLBG	513	509	0.99	97.66%
LPPR	513	508	0.99	98.83%
LROP	504	493	0.98	97.82%
LEIB	495	490	0.99	98.99%
EGBB	474	468	0.99	98.73%
LIML	458	452	0.99	98.47%
LIME	453	450	0.99	99.34%
LIRN	449	337	0.75	74.61%
LFML	444	440	0.99	98.87%
EDDS	440	418	0.95	94.32%
LGIR	424	1	0.00	0.24%
LIPZ	412	409	0.99	99.03%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LFLL	411	417	1.01	97.81%
LPFR	405	399	0.99	98.52%
LYBE	394	391	0.99	98.73%
LICC	390	382	0.98	97.69%
LTAC	381	0	0.00	0.00%
LIPE	380	375	0.99	98.68%
LCLK	368	352	0.96	95.38%
EPKK	360	348	0.97	96.39%
GMMN	360	387	1.07	87.78%
EGGD	359	356	0.99	98.89%
LEVC	355	341	0.96	96.06%
LFSB	354	354	1.00	98.59%
ENBR	349	366	1.05	98.28%
LTBJ	346	275	0.79	79.48%
EGPF	344	328	0.95	93.60%
LGRP	340	334	0.98	95.88%
LMML	326	325	1.00	99.39%
GCTS	321	11	0.03	0.00%
GCXO	319	367	1.15	96.24%
LATI	316	0	0.00	0.00%
LICJ	310	288	0.93	92.58%
EBCI	304	304	1.00	100.00%
BIKF	297	331	1.11	98.99%
ELLX	294	292	0.99	99.32%
LFBO	293	291	0.99	98.63%
LGTS	279	271	0.97	96.77%
GMMX	278	88	0.32	30.94%
LEZL	275	273	0.99	99.27%
EVRA	273	271	0.99	98.90%
LTBS	273	219	0.80	80.22%
LBSF	272	169	0.62	62.13%
GCRR	269	0	0.00	0.00%
EGNX	267	265	0.99	98.50%
EPKT	259	249	0.96	96.14%
HEGN	257	0	0.00	0.00%
UGTB	257	199	0.77	61.87%
LFRS	256	254	0.99	98.83%
EDDP	256	257	1.00	99.61%
EPGD	248	39	0.16	15.73%
LGKR	244	212	0.87	84.84%
HECA	244	42	0.17	17.21%
EGAA	239	240	1.00	100.00%
ENZV	236	174	0.74	67.80%
LEBB	234	174	0.74	73.50%
LFPB	230	230	1.00	98.70%
LIEO	228	195	0.86	85.53%
LIBD	228	0	0.00	0.00%

Table 9: The 100 busiest aerodromes by Network Manager arrivals, against the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Out-of-area is excluded – it is not an aerodrome and is reported separately in Section 7. *Count ratio* is OPDI movements divided by Network Manager movements, so it can sit near 1 even when the individual flights are wrong; *recall* is the share of Network Manager arrivals OPDI put at this aerodrome, and is the stricter measure.

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LTFM	2,258	1,647	0.73	72.85%
EHAM	2,136	1,766	0.83	82.49%
EDDF	2,091	1,609	0.77	76.66%
LFPG	2,066	1,546	0.75	74.54%
EGLL	1,986	1,183	0.60	57.85%
LEMD	1,786	1,357	0.76	74.92%
LEBL	1,597	1,411	0.88	88.17%
EDDM	1,571	1,313	0.84	83.45%
LTAI	1,437	994	0.69	68.89%
LEPA	1,436	1,386	0.97	96.24%
LIRF	1,433	1,189	0.83	82.76%
LGAV	1,317	1,128	0.86	83.45%
EGKK	1,220	994	0.81	81.39%
EIDW	1,188	965	0.81	81.14%
LSZH	1,160	978	0.84	83.97%
EKCH	1,152	1,007	0.87	87.41%
LOWW	1,145	896	0.78	78.17%
LTFJ	1,138	925	0.81	80.67%
LPPT	1,001	764	0.76	76.32%
LFPO	991	888	0.90	89.51%
LIMC	967	757	0.78	77.56%
ENGM	940	877	0.93	92.66%
EGCC	924	619	0.67	66.99%
EGSS	888	773	0.87	86.49%
EPWA	880	705	0.80	79.77%
EBBR	871	733	0.84	83.47%
LEMG	861	789	0.92	91.64%
ESSA	841	709	0.84	83.83%
EDDB	840	763	0.91	89.88%
EDDL	751	666	0.89	88.15%
LFMN	710	657	0.93	91.97%
LSGG	670	625	0.93	91.64%
EFHK	666	577	0.87	85.59%
LKPR	657	571	0.87	86.45%
LEAL	599	572	0.95	95.49%
EGPH	584	506	0.87	85.96%
LHBP	577	238	0.41	41.07%
EGGW	552	479	0.87	86.41%
EDDH	532	488	0.92	90.04%
GCLP	521	0	0.00	0.00%
LLBG	516	409	0.79	74.03%
LPPR	512	435	0.85	84.77%
EDDK	509	443	0.87	85.66%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LEIB	509	493	0.97	96.66%
LROP	509	463	0.91	90.77%
EGBB	466	388	0.83	83.26%
LIML	465	445	0.96	94.62%
LIME	460	419	0.91	91.09%
LIRN	447	335	0.75	73.60%
LFML	445	391	0.88	87.64%
EDDS	433	377	0.87	86.84%
LGIR	430	0	0.00	0.00%
LFLL	412	386	0.94	90.53%
LIPZ	409	370	0.90	89.98%
LPFR	407	392	0.96	96.31%
LYBE	395	342	0.87	86.33%
LICC	391	356	0.91	91.05%
LIPE	376	359	0.95	95.21%
LTAC	374	1	0.00	0.00%
LCLK	367	338	0.92	91.55%
EGGD	360	303	0.84	84.17%
ENBR	357	356	1.00	93.00%
GMMN	357	336	0.94	71.99%
LEVC	357	333	0.93	93.00%
EPKK	356	250	0.70	70.22%
LFSB	352	322	0.91	91.48%
LGRP	346	312	0.90	87.57%
EGPF	346	299	0.86	85.55%
LTBJ	341	275	0.81	80.65%
LMML	326	310	0.95	94.79%
GCTS	323	97	0.30	21.98%
LATI	323	0	0.00	0.00%
GCXO	320	316	0.99	23.75%
LICJ	314	291	0.93	92.04%
EBCI	303	285	0.94	94.06%
BIKF	296	165	0.56	29.39%
ELLX	294	226	0.77	76.53%
LFBO	290	278	0.96	95.17%
LGTS	281	264	0.94	93.95%
LTBS	280	228	0.81	81.07%
GMMX	279	57	0.20	19.35%
EVRA	278	221	0.79	78.78%
LEZL	276	255	0.92	92.39%
GCRR	269	0	0.00	0.00%
LBSF	267	89	0.33	32.58%
EDDP	261	224	0.86	84.67%
HEGN	258	0	0.00	0.00%
LFRS	257	237	0.92	92.22%
EPKT	257	211	0.82	82.10%
UGTB	257	101	0.39	21.79%
EGNX	255	206	0.81	79.61%
LIEO	252	212	0.84	83.33%
LGKR	247	222	0.90	68.02%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
EGAA	246	223	0.91	90.65%
EPGD	244	10	0.04	1.64%
ENZV	236	130	0.55	46.61%
LEBB	235	53	0.23	16.17%
LTFE	232	187	0.81	80.17%
HECA	231	82	0.35	34.20%
LIBD	229	1	0.00	0.00%

The error is an **undercount** rather than an overcount, which is the correctable direction: a systematic shortfall can be adjusted for, an inflated count cannot. Roughly six points of it are the out-of-area bucket rather than per-aerodrome detection failure — flights the method never labelled land nowhere, so they show up as missing movements at whichever aerodrome the truth names.

Recall also scales sharply with aerodrome size. Across the hundred busiest — 74.96% of all departures in the sample — the median is 98.24% and the quartiles are 95.60% and 99.04%, against a median of 80.00% over all 824 aerodromes with departures. The flight list is therefore substantially more complete at large aerodromes than small ones, which matters before comparing across aerodrome classes — and Section 8.1 says why.

8.2 Busy aerodromes that fail outright

Extending the listing from thirty rows to a hundred exposes something the shorter view could not: a handful of genuinely busy aerodromes where the method does not merely lose a few movements but fails almost completely.

29 of the two hundred rows above sit below 50% recall — 11 departure aerodromes and 18 arrival ones. They split cleanly into two kinds of failure:

- **Never seen** (19 rows). Under a quarter of their flights get an answer at all: Gran Canaria departures return 24 answers against 520 movements, and Hurghada none at all. These cluster where ADS-B receiver coverage is thin — the Canaries, North Africa, the Caucasus, Anatolia. No detection rule can recover a flight with no low-altitude trajectory.
- **Seen and misnamed** (10 rows). The trajectory is there and the answer is wrong. Bilbao arrivals are answered 90.64% of the time and only 38 of 235 land on Bilbao.

Tenerife North is the cleanest warning about reading count ratios. It is assigned 316 arrivals against 320 true ones — a count ratio of 0.99, which looks like a well-measured aerodrome — while its recall is 23.75%. Almost every movement it is credited with belongs somewhere else, and the totals cancel the error exactly as Section 8 warned they could.

Every aerodrome in both groups is a `large_airport` with scheduled service and is present in the detection reference, so neither failure is the reference gap of Section 8.1. The first group is a data limitation and is not fixable here. The second is a detection failure at aerodromes large enough to matter, it is concentrated rather than scattered, and **this study has not diagnosed it** — it is the most promising lead the cross-check produces.

9 Choosing the operating point

The sections above establish how each parameter behaves. This one picks values, and argues for them in flights rather than in percentages — because a rate conceals the size of what is being traded. "−0.84 pp of accuracy" and "792 more wrong aerodromes in three days" are the same fact, and only one of them can be weighed against anything.

All counts are out of 95,116 ground-truth flights per role, over 2025-06-05 to 07. *Answered* is what the method labelled, *correct* is what it labelled right, *wrong* is what it labelled wrong, *silent* is what it declined. The last three columns compare against production (`trend`).

Table 10: Departures: the parameter grid in rates and in flights, against production. Penalty fixed at 10 NM. *Gain:err* is extra correct labels per extra wrong label, against production.

Parameters	Cov	Acc	Ovr	Answered	Correct	Wrong	Silent	d correct	d wrong	Gain:err
trend (production)	68.72%	98.82%	67.91%	65,365	64,594	771	29,751	—	—	—
20 NM / 15,000 ft	80.07%	98.44%	78.82%	76,164	74,973	1,191	18,952	10,379	420	24.7:1
30 NM / 15,000 ft	80.96%	98.16%	79.47%	77,003	75,587	1,416	18,113	10,993	645	17.0:1
40 NM / 15,000 ft	81.20%	97.98%	79.55%	77,230	75,667	1,563	17,886	11,073	792	14.0:1
20 NM / 20,000 ft	80.53%	98.11%	79.01%	76,593	75,148	1,445	18,523	10,554	674	15.7:1
30 NM / 20,000 ft	81.86%	97.65%	79.94%	77,863	76,032	1,831	17,253	11,438	1,060	10.8:1
40 NM / 20,000 ft	82.44%	97.34%	80.24%	78,411	76,322	2,089	16,705	11,728	1,318	8.9:1
20 NM / 25,000 ft	80.80%	97.80%	79.02%	76,852	75,162	1,690	18,264	10,568	919	11.5:1
30 NM / 25,000 ft	82.41%	97.06%	79.98%	78,382	76,078	2,304	16,734	11,484	1,533	7.5:1
40 NM / 25,000 ft	83.50%	96.31%	80.41%	79,419	76,487	2,932	15,697	11,893	2,161	5.5:1
20 NM / 30,000 ft	81.13%	97.40%	79.02%	77,168	75,165	2,003	17,948	10,571	1,232	8.6:1
30 NM / 30,000 ft	83.09%	96.26%	79.99%	79,035	76,083	2,952	16,081	11,489	2,181	5.3:1
40 NM / 30,000 ft	84.44%	95.21%	80.39%	80,317	76,467	3,850	14,799	11,873	3,079	3.9:1

Table 11: Arrivals: the same grid. Note that most cells produce *fewer* correct labels than production while adding wrong ones. Penalty fixed at 10 NM.

Parameters	Cov	Acc	Ovr	Answered	Correct	Wrong	Silent	d correct	d wrong	Gain:err
trend (production)	68.06%	97.87%	66.61%	64,738	63,361	1,377	30,378	—	—	—
20 NM / 15,000 ft	66.78%	96.65%	64.54%	63,514	61,387	2,127	31,602	-1,974	750	—
30 NM / 15,000 ft	69.10%	95.63%	66.08%	65,726	62,856	2,870	29,390	-505	1,493	—

Parameters	Cov	Acc	Ovr	Answered	Correct	Wrong	Silent	d correct	d wrong	Gain:err
40 NM / 15,000 ft	69.97%	95.10%	66.54%	66,551	63,287	3,264	28,565	-74	1,887	—
20 NM / 20,000 ft	67.42%	95.97%	64.70%	64,124	61,543	2,581	30,992	-1,818	1,204	—
30 NM / 20,000 ft	70.32%	94.24%	66.27%	66,883	63,032	3,851	28,233	-329	2,474	—
40 NM / 20,000 ft	71.47%	93.47%	66.81%	67,983	63,544	4,439	27,133	183	3,062	0.1:1
20 NM / 25,000 ft	67.91%	95.54%	64.88%	64,592	61,710	2,882	30,524	-1,651	1,505	—
30 NM / 25,000 ft	71.09%	93.46%	66.45%	67,622	63,202	4,420	27,494	-159	3,043	—
40 NM / 25,000 ft	72.43%	92.51%	67.00%	68,889	63,728	5,161	26,227	367	3,784	0.1:1
20 NM / 30,000 ft	68.50%	95.04%	65.10%	65,156	61,923	3,233	29,960	-1,438	1,856	—
30 NM / 30,000 ft	71.96%	92.65%	66.67%	68,445	63,417	5,028	26,671	56	3,651	0.0:1
40 NM / 30,000 ft	73.55%	91.40%	67.23%	69,962	63,946	6,016	25,154	585	4,639	0.1:1

9.1 Departures: every cell wins, but the marginal return collapses

Every cell in the table beats production. That is not the question. The question is what each successive loosening *buys*, and the answer is that it collapses.

Table 12: Departures: each step of loosening, and its marginal return. *Marginal* is extra correct labels per extra wrong label for that step alone, not against production.

Step	d correct	d wrong	Marginal
production to 20 NM / 5,000 ft	5,366	26	206.38:1
20 NM / 5,000 ft to 20 NM / 8,000 ft	2,426	62	39.13:1
20 NM / 8,000 ft to 20 NM / 10,000 ft	1,295	94	13.78:1
20 NM / 10,000 ft to 20 NM / 15,000 ft	1,292	238	5.43:1
20 NM / 15,000 ft to 30 NM / 15,000 ft	614	225	2.73:1
30 NM / 15,000 ft to 40 NM / 15,000 ft	80	147	0.54:1
40 NM / 15,000 ft to 40 NM / 20,000 ft	655	526	1.25:1
40 NM / 20,000 ft to 40 NM / 25,000 ft	165	843	0.20:1
40 NM / 25,000 ft to 40 NM / 30,000 ft	-20	918	loss

Two readings of the same table. Against production, every cell looks excellent — the worst ratio in the grid is still better than three correct labels per error. Step by step, the picture inverts: the first move off production is nearly free, and by the wide-and-high end each additional correct label costs more than one additional error.

Note where the currently published operating point sits. Going from 30 NM to 40 NM at 15,000 ft adds 80 correct labels and 147 wrong ones. **The published parameters are already past break-even**, which the percentage view never showed because it is a 0.18 pp change in accuracy.

9.2 Arrivals: nothing in the grid earns its place

Arrivals invert the departure result completely. **8 of the 12 cells produce fewer correct arrival labels than production**, while every one of them produces more wrong ones. The cells that do gain correct labels buy them at a ruinous rate:

Table 13: Arrivals: what each cell costs against production. Cells omitted from the ratio column produce fewer correct labels than production and so gain nothing to price.

Cell	d correct	d wrong	Errors per correct gained
30 NM / 15,000 ft	-505	1,493	gains none
40 NM / 15,000 ft	-74	1,887	gains none
40 NM / 20,000 ft	183	3,062	16.7
40 NM / 25,000 ft	367	3,784	10.3
40 NM / 30,000 ft	585	4,639	7.9

The best arrival cell in the entire grid, 40 NM / 30,000 ft, gains 585 correct labels and adds 4,639 wrong ones. There is no exchange rate under which that is a good trade. **`endpoint` should not replace `trend` for arrivals at any setting measured here** — and Section 10 asks why, and what could be done about it.

9.3 Why no single metric can make this choice

A natural instinct is to let one number decide. It cannot, and it is worth seeing why before trusting any headline figure.

Table 14: Departures: the best cell in the whole grid under each metric taken alone.

Optimising	Parameters	Coverage	Accuracy	Overall
coverage	110 NM / unrestricted	91.41%	87.91%	80.36%
accuracy	5 NM / 2,000 ft	66.84%	99.24%	66.33%
overall	80 NM / 30,000 ft	85.63%	94.19%	80.66%
—	trend (production)	68.72%	98.82%	67.91%
—	nearest (no abstention)	91.68%	87.78%	80.48%

Table 15: Arrivals: the same. Note that no `endpoint` cell reaches `nearest`'s overall correctness.

Optimising	Parameters	Coverage	Accuracy	Overall
coverage	110 NM / unrestricted	87.75%	79.53%	69.79%
accuracy	5 NM / 500 ft	52.03%	99.15%	51.59%
overall	100 NM / unrestricted	87.44%	79.82%	69.79%
—	trend (production)	68.06%	97.87%	66.61%
—	nearest (no abstention)	88.56%	79.41%	70.33%

Each of these is degenerate in its own way.

Maximising accuracy produces a rule quieter than the one it replaces. The best-accuracy departure cell answers 66.84% of flights against production's 68.72%. It is more accurate and *less useful*: fewer flights labelled, for a fraction of a point. For arrivals it is starker still, at 52.03% coverage.

Maximising coverage switches abstention off. The best-coverage cell reaches 91.41% against `nearest`'s 91.68% — the residual is only the aerodromes with no elevation in OurAirports. At that setting `endpoint` *is* `nearest`, and the reason for choosing it has been configured away.

Maximising overall correctness is maximising coverage in disguise. Across the grid, overall correctness correlates **+0.988 with coverage** and **-0.439 with accuracy** for departures (+0.951 and -0.584 for arrivals). Coverage spans 60.27% across the grid and accuracy only 11.33%, so the product tracks the wider factor. And for arrivals the verdict of overall correctness is explicitly *do not abstain*: `nearest` scores 70.33%, above every `endpoint` cell.

It is also too flat to discriminate. 30 of the 126 departure cells sit within half a point of the best overall correctness, spanning 40–110 NM and 20,000–40,000 ft. Those statistically indistinguishable cells differ by **thousands of wrong labels**. Overall correctness cannot see the thing that separates them.

9.4 The exchange rate, and the choice

What remains is a policy question that the data can inform but not settle: how many extra wrong labels is one extra correct label worth?

There is a structural asymmetry that argues for pricing errors above correct answers. **A wrong aerodrome corrupts two movement counts** — the aerodrome that loses the flight and the aerodrome that wrongly gains it — and it is invisible to a consumer, indistinguishable from a correct label. **A silent flight depresses one count and is explicit**: it arrives as a null the consumer can filter, and Section 8's undercount is correctable in a way an inflated count is not. Tenerife North makes the point concretely: a count ratio of 0.99 that looks healthy, on a recall of 23.75%.

On that reasoning an error is worth at least two correct labels, and the departure ladder stops at the last step above 2:1.

The operating point this study recommends

Departures: `endpoint` at 30 NM, 15,000 ft, 10 NM scheduled-service penalty. 10,993 correct departure labels against production for 645 wrong ones.
Arrivals: keep `trend`. Nothing measured here improves on it, and the cells that add correct labels add errors seven to seventeen times faster.
At a stricter exchange rate the departure answer moves inward — 5:1 gives 20 NM / 10,000 ft, 10:1 gives 20 NM / 8,000 ft — and at 1:1 it moves out to 40 NM / 20,000 ft. The rate is the decision; the parameters follow from it.

The penalty of 10 NM is carried over from Section 6, where it was swept at 40 NM and 15,000 ft. It has not been re-verified at 30 NM, and should be before this is published.

10 What `endpoint` could borrow from `trend`

Section 9.2 leaves an uncomfortable result: for arrivals, the algorithm this study has been arguing for loses to the one it proposed to replace, at every setting measured. That is worth understanding rather than working around, because the reason points at what a better rule would look like.

10.1 Why `trend` wins arrivals

The two rules differ in how much of the trajectory they read.

`endpoint` reads one sample. The first or last fix of the track, its distance to each candidate, its height above field elevation, its ground flag. Nothing else.

`trend` reads every sample below FL40 inside the zone. For each (track, aerodrome) pair it smooths altitude over five samples and counts rises against falls. It sees a descent, not a point.

For departures that difference barely matters. A departure's first sample is near the field whatever else is true, so one sample locates it almost as well as fifty — which is why `endpoint` beats `trend` on departures by 10,993 correct labels.

For arrivals it matters a great deal, for the reason Section 5 identified: **an arrival's last sample is where reception died, which is a property of the receiver network rather than of the flight.** The samples *before* it still describe an approach. `endpoint` throws them away; `trend` uses them.

There is a second difference, and it is the more important one. `trend`'s altitude vote is evaluated **per candidate aerodrome** — it asks "is this aircraft descending toward *this* aerodrome", which discriminates between candidates. `endpoint`'s thresholds are a **gate**: as Section 3 puts it, the abstention decides *whether* to answer, never *what* to answer. Naming is left entirely to effective distance.

That matters because the dominant arrival failure is misnaming, not silence. Section 8.2 found Bilbao answering 90.64% of its arrivals and placing only 38 of 235 correctly. **No gate can fix a naming error.** Tightening the thresholds makes the rule answer less; it does not make it answer better.

10.2 Tested: an approach cone instead of a rectangle

The first idea to try is the cheapest, because it needs no new data — only a different shape for the same two numbers.

The current rule accepts an endpoint inside `radius_nm` **and** below `height_ft`, treating distance and height as independent. On an approach they are not: an aircraft descending toward an aerodrome trades height for distance at roughly a constant rate. The 3° glideslope is 318 ft/NM and the "three times the flight level" descent rule of thumb is about 333 ft/NM. So the endpoints that genuinely belong to an aerodrome should lie in a **cone**, `agl_ft` `k` · `dist_nm`, not a box.

The two shapes disagree in both corners:

- **close and high** is inside the box, outside the cone. An aircraft at 15,000 ft overhead an aerodrome is overflying it. The box accepts this, and it is a pure source of wrong labels.

- **far and low** is outside the box, inside the cone. An arrival whose reception dies at 8,000 ft and 30 NM is on profile and is a good candidate.

Table 16: The approach cone at a 40 NM outer cap, against the rectangle **at matched coverage** – the only fair comparison between two acceptance shapes. *box acc* is the rectangle's accuracy interpolated to the cone's coverage; *cone gain* is positive where the cone is better.

Slope	ADEP cov	ADEP acc	box acc	cone gain	ADES cov	ADES acc	box acc	cone gain
150 ft/NM	56.76%	98.81%	98.94%	-0.13 pp	57.34%	98.22%	98.48%	-0.26 pp
200 ft/NM	56.80%	98.75%	98.94%	-0.19 pp	57.60%	98.12%	98.46%	-0.34 pp
250 ft/NM	56.87%	98.65%	98.94%	-0.29 pp	58.89%	98.01%	98.39%	-0.38 pp
300 ft/NM	57.01%	98.49%	98.94%	-0.45 pp	62.09%	97.90%	98.05%	-0.15 pp
333 ft/NM	57.16%	98.36%	98.94%	-0.58 pp	63.37%	97.80%	97.89%	-0.09 pp
400 ft/NM	57.67%	98.29%	98.94%	-0.65 pp	65.22%	97.56%	97.42%	+0.14 pp
500 ft/NM	60.06%	98.12%	98.95%	-0.83 pp	66.71%	96.93%	96.92%	+0.02 pp
650 ft/NM	66.92%	97.59%	98.98%	-1.39 pp	68.14%	95.79%	96.24%	-0.45 pp
800 ft/NM	73.44%	97.06%	98.77%	-1.72 pp	69.34%	94.62%	95.49%	-0.87 pp
1000 ft/NM	78.92%	95.86%	98.41%	-2.55 pp	70.97%	92.72%	94.01%	-1.29 pp
1500 ft/NM	84.45%	93.65%	95.20%	-1.55 pp	74.06%	89.32%	90.94%	-1.62 pp

It does not work. For departures the cone is worse at every slope — by up to 2.55 pp of accuracy at matched coverage. For arrivals it is a wash: negative at most slopes, and the largest positive figure is a tenth of a point, which is noise on this sample.

The departure failure is physical and worth stating, because it explains why the box is the right shape there. **Climb gradients are far steeper than descent gradients.** A departure 10 NM out is routinely at 6,000–8,000 ft — 600 to 800 ft/NM, against the glideslope's 318 — so a cone tuned to descent excludes precisely the departures it should keep. The box's permissiveness close to the field is not an oversight; it is what departures need. To reach the coverage the box gives at 15,000 ft, the cone needs a slope above 1,000 ft/NM, at which point it is no longer a cone in any meaningful sense.

! The negative result that matters most

Neither shape closes the arrival gap. At the coverage where production sits, the cone reaches 95.79% accuracy against `trend`'s 97.87%.
Box and cone are both **acceptance regions defined on the distance and height of a single endpoint**, and this experiment suggests that family is exhausted. No rearrange-

ment of those two numbers recovers what `trend` has. The remaining gains have to come from somewhere else: **more evidence**, or **a naming rule**.

10.3 Untested, in order of expected value

1. A descent-direction tie-break — a naming rule, not a gate. Rank candidates by effective distance as now, but when the top two are within a small margin, break the tie on which aerodrome the aircraft is descending *toward*, using the vertical rate at the endpoint and the bearing to each candidate. This is the smallest piece of `trend` that can be borrowed, and it is the only proposal here that attacks misnaming. It fires only on the ambiguous pairs, so it cannot damage the flights that are already unambiguous — which is exactly the property Section 6 found made the scheduled-service penalty safe. The candidate cache would need `vert_rate` and a bearing, both available in the state vectors at no join cost.

2. Closest approach below FL40, instead of the last sample. Replace "the last fix" with "the fix of closest approach among those below FL40". For an arrival whose reception fails on the approach, the closest approach is a far better locator than the last recorded point. The cost is real: the candidate builder currently emits two rows per track, which is what makes the sweeps in this paper affordable (Section 1), and this would emit more. A middle course is to keep the two endpoints and add the closest-approach fix as a third.

3. Re-testing the cascade — but read version 1 first. The obvious move is to take `trend`'s answer when it has one and fall through to `endpoint`. **Version 1 tested exactly this and it failed**, and the way it failed is the caution: every rung had *negative lift*, meaning the trend rungs fired on easy flights where nearest was already right, and overwrote better answers. The cascade collapsed onto its fallback rung.

Whether that still holds is an open question rather than a settled one, because version 1 measured it in the pre-H3 architecture and on departures, and this version's arrival result is new. The diagnostic is cheap and decisive: attribute each cascade answer to the rung that supplied it, and score that rung against **`endpoint` on the same flights**. If `trend`'s arrival advantage comes from answering *different* flights, a cascade will help; if it comes from being better on *shared* flights, `trend` should simply replace `endpoint` for arrivals, which is what Section 9.2 already recommends. Version 1's harness has the attribution code (`--diagnose-cascade`).

> ⚠ Not this: extrapolating the trajectory
>
> The intuitive fix for an arrival whose reception dies at 8,000 ft is to project the descent forward to the ground. **Do not.** Version 2 of this study implemented trajectory extrapolation and it failed, and the 2025 PRC Data Challenge reached the same conclusion independently and more strongly: *"attempts to complete the trajectory always lead to worse result"*. Two independent negative results on the same idea is enough.

10.4 Should any of this apply to departures too?

The tie-break of proposal 1 should, and for the same reason it helps arrivals: Section 6 showed the largest identified departure error class was a neighbouring-aerodrome confusion, and the scheduled-service penalty only fixed the subset where the neighbour was military. A descent-direction test — inverted, a *climb*-direction test — discriminates on physics rather than on aerodrome metadata, and would catch confusions the penalty cannot.

Proposals 2 and 3 should not. Departures do not have the evidence problem: one sample already locates them, `endpoint` already beats `trend` there by 10,993 correct labels, and adding trajectory evidence to a rule that is not short of evidence is how the version 1 cascade lost accuracy.

11 Recommendation

Adopt `endpoint` for departures if coverage matters more than the last point of accuracy. It adds +12.47 pp of coverage for -0.84 pp, and it is a smaller algorithm than the one it replaces — no smoothing window, no margin parameter, no per-aerodrome vote. If accuracy is paramount, keep `trend`; it is genuinely the more accurate rule now.

Use 20–40 NM and a scheduled-service penalty of 10 NM. Both are on plateaus, so neither is delicate.

Do not change the arrivals algorithm on this evidence. Arrivals are weaker on every measure and every version of this study has found the same.

Publish the provenance column. `adep_source` / `ades_source` distinguish "named an aerodrome", "outside the observed area" and "could not determine". Today all three are a single null in the published flight list, and they mean entirely different things to a consumer. This is the highest-value change per line of code in the whole study.

Make abstention aware of what the reference contains. Flights to and from small aerodromes and heliports are answered about half the time and are wrong every time (Section 8.1). They are a small share of movements but they are false statements rather than gaps, and they are indistinguishable in the output from correct answers. Either extend the reference to small aerodromes or abstain when the nearest candidate is far enough away to suggest the real aerodrome is one the reference does not hold.

12 Limitations

- **Three days**, one sample. Versions 2 and 3 used two samples a year apart; this version re-runs the whole pipeline, which is expensive, so it uses one. The mode ranking is robust across earlier versions, but the exact figures here have not been validated on a second period.
- **Out-of-area recall is low.** The geometric border test is the wrong instrument; the phase-based test in Section 7 is the successor.
- **The detection reference holds large and medium aerodromes only**, so accuracy on any other class is zero by construction rather than by measurement. The per-class table quantifies this; it does not fix it.
- **The scheduled-service preference has a cost** that these accuracy figures include but do not itemise: movements at non-scheduled aerodromes are pulled to nearby hubs.
- **The sweep re-implements the endpoint rule** as a DataFrame operation instead of calling the pipeline's `classify_endpoints`, because a sweep cell needs only `(track_id, adep, ades)` while the pipeline method builds a whole flight list. The headline `endpoint` row *does* come from the pipeline path, so a divergence between the two would surface as the sweep disagreeing with its own mode — but this is a duplication and worth removing.
- **`track_id` is taken as given**, with the fragmentation and merging rates version 2 measured.

13 Reproducing

Reference zones, state vectors and tracks, in the `opdi` repository:

```
RANGE="--env opensky --start 2025-06-05 --end 2025-06-08"
opdi run $RANGE --step 00a   # zones: 5 NM bands to 110 NM
opdi run $RANGE --step 01    # state vectors, bbox and 5 s applied at ingest
opdi run $RANGE --step 02    # tracks, with H3 res 7 and 12
```

Candidates once, then a flight list per mode:

```
from datetime import date
from opdi.pipeline.flights import FlightListProcessor

fp = FlightListProcessor(spark, config)
fp.build_endpoint_candidates(date(2025, 6, 1))
for mode in ("trend", "nearest", "endpoint"):
    fp.process_dai(date(2025, 6, 1), mode=mode, skip_if_processed=False,
                   abstention_radius_nm=40, abstention_height_ft=15000,
                   sched_penalty_nm=10,
                   table_name=f"research/flight_list_{mode}",
                   write_mode="overwrite")
```

Scoring and both sweeps. This runs locally: the inputs are small once the candidates are cached, and the cluster namespace is shared.

```
python benchmarks/benchmark_modes.py --months 202506 \
    --days 2025-06-05 2025-06-06 2025-06-07 --local --results-dir <dir>
```

Rendering this page needs no credentials, no database and no cluster.